

RAMALINGAM JAYAVELU

Software Engineering Student · Backend Engineering · System Design

Chennai, Tamil Nadu | contact@linga.engineer | [Github](#) | [Portfolio](#) | [Leetcode](#) | [Engineering Blogs](#) | [LinkedIn](#)

SUMMARY

Final-year Robotics & Automation Engineering student with a Minor in Computer Science & Engineering, focused on backend engineering and software architecture. Designs software by understanding problem constraints, data ownership, and domain relationships before selecting technologies. Experience building scalable backend systems, AI retrieval pipelines, and domain-driven software systems using Java, Spring Boot, Python, and JavaScript.

PROJECTS

Learn With Linga | [React](#), [Vite](#), [JavaScript](#), [Serverless Functions](#)

github.com/lingarobotics/learning-insights-lgc

Live Demo: [insights.linga.engineer](#) — Deployed in Vercel

- Architected a convention-over-configuration publishing platform where a single Markdown file serves as the source of truth for content, metadata, and route generation — currently publishing **50+ engineering insights across 11 domains** without requiring application code changes.
- Implemented filesystem-backed serverless APIs that dynamically resolve domain routes and expose Markdown content through structured JSON responses, using the filesystem as the **authoritative content index**.
- Designed a zero-configuration publishing pipeline where convention-based Markdown files automatically trigger content discovery, metadata derivation, and dynamic content rendering.

BugSense AI | [FastAPI](#), [FAISS](#), [Sentence Transformers](#), [React](#), [Vite](#), [Python](#)

github.com/lingarobotics/bug-sense-ai

- Initiated and led a hackathon team project, defining a modular software architecture across AI, service, and API layers while coordinating feature ownership and integrating frontend and backend components for end-to-end system delivery.
- Designed and implemented a semantic bug retrieval pipeline using the **all-MiniLM-L6-v2 Sentence Transformer** to generate **384-dimensional embeddings**, indexing **15,000 processed bug reports** in a FAISS vector store with threshold-based similarity classification for duplicate, possible-duplicate, and new defect detection.
- Implemented cosine-based DBSCAN clustering on semantic embeddings to group related defect reports, improving repository-level duplicate analysis across an 88,000+ report source dataset.

ONGOING ENGINEERING PROJECTS

ResultGrid | [Java](#), [Spring Boot](#), [PostgreSQL](#)

github.com/lingarobotics/ResultGrid

- Designing a high-read backend platform targeting support for **~18,000 concurrent client requests** for results during university result publication periods, modeling academic data as a relational hierarchy (District → College → Department → Student → Result) to enable predictable retrieval and scalable API design.
- Architected a layered Spring Boot backend with verification, controller, service, caching, and persistence layers; evaluated **Caffeine caching** and database connection pooling strategies to support high volumes of concurrent client requests while **improving observability with Spring Boot Actuator**.
- Evaluating delivery architectures for immutable examination results by comparing request-driven retrieval with **pre-generated result artifacts**, balancing implementation complexity, scalability, and reliability for burst traffic scenarios.

MathLogic | [React](#), [Vite](#), [Supabase \(PostgreSQL\)](#)

github.com/lingarobotics/MathLogic

- Designing a concept-centered content model for **R2021 MA3151 Unit I (8 mathematical topics)**, where reusable learning workflows are derived from problem-solving methods and concepts remain independent of curriculum organization through regulation-based mappings, enabling reuse across multiple academic regulations without duplication.
- Modeling each topic by deriving reusable learning workflows through **pattern extraction** from worked examples, enabling step-level progress tracking and concept-level comprehension.

Codebase Insight | [React](#), [Supabase \(PostgreSQL\)](#)

github.com/lingarobotics/CodeBase-Insight

- Architecting a content-driven repository learning platform where repository architecture, modules, relationships, and engineering decisions are modeled as **structured lesson content** and rendered through reusable frontend components, enabling new repositories to be onboarded through content authoring while preserving the same rendering architecture.
- Designed a relational schema in Supabase to capture structured learning signals (**Understood, Not Yet Understood, Valuable, Design Reasoning Understood**), enabling learner **engagement tracking and progress analysis** across repository content.

EDUCATION

B.E. Robotics & Automation Engineering | Minor: Computer Science and Engineering (Full Stack Development)

CGPA: 7.85

Karpaga Vinayaga College of Engineering and Technology

2023-27

CERTIFICATIONS

Oracle Agentic AI Certified Foundations Associate | Sep 2026 [View E-Certificate](#) | [View Docs](#) | [Verify Credential Here](#)
Oracle Cloud Infrastructure AI Foundations Associate | Sep 2026 [View E-Certificate](#) | [View Docs](#) | [Verify Credential Here](#)

TECHNICAL SKILLS

Languages & Fundamentals: Java, Python, JavaScript, TypeScript, SQL, Data Structures & Algorithms

Frameworks & Backend: Spring Boot, Spring Security, JWT, Node.js, Express.js, REST APIs, Serverless Functions, React.js

Databases: PostgreSQL, MongoDB

Tools, Cloud & Platforms: Git, GitHub, Postman, IntelliJ IDEA, VS Code, Supabase, Vercel

AI & Machine Learning: ML, Deep Learning, GenAI, LLMs, Agentic AI, LangChain, OpenAI SDK, Agentic Orchestration